Dirac-Monopole Dynamics on the Sphere: Magnetic Circles and the Complete Covariant-Laplacian Spectrum

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Abstract

We identify a conserved Poincaré vector and every magnetic circle on the unit sphere, including its least period and all zero-energy and zero-charge faces. This is the *Poincare vector and every magnetic circle* stage. For an integral charge we then derive the *complete monopole-harmonic spectrum*, its multiplicities, heat trace, and charge-conjugation symmetry.

1 The classical flow

Let $S^2 \subset \mathbb{R}^3$ be the oriented unit sphere and let $Jv = x \times v$ on T_xS^2. Fix $q \in \mathbb{Z}$, put $b = q/2$, and consider

$$\nabla_t \dot{x} = bJ\dot{x}, \qquad |\dot{x}|^2 = 2E. \tag{1}$$

Theorem 1 (All magnetic trajectories)**.** *The vector*

$$K = x \times \dot{x} + bx$$

is constant and satisfies

$$K \cdot x = b, \qquad |K|^2 = 2E + b^2.$$

If $E > 0$, the trajectory is the oriented small circle cut out by $K \cdot x = b$ and its least positive period is

$$T(E, q) = \frac{2\pi}{\sqrt{2E + q^2/4}}. \tag{2}$$

If $E = 0$, the trajectory is stationary. If $q = 0$ and $E > 0$, the circle is a great circle.

Proof. Skew-symmetry of J preserves $|\dot{x}|$. The ambient acceleration is $\ddot{x} = b\,x \times \dot{x} - 2Ex$. Consequently,

$$\dot{K} = x \times \ddot{x} + b\dot{x} = b\,x \times (x \times \dot{x}) + b\dot{x} = 0.$$

Orthogonality gives the two displayed identities. The triple-product identity also gives $K \times x = \dot{x}$, so x is a rigid rotation about the fixed axis K with angular speed $|K|$. Its squared circle radius is $2E/(2E + b^2)$, positive exactly when $E > 0$. A nonconstant circle first returns with its tangent after angle 2π, proving (2). The two boundary statements follow directly. □

Time reversal gives the precise sign pairing: if $x_q(t)$ solves charge q, then $x_q(-t)$, with its initial velocity reversed, solves charge $-q$ and traverses the same geometric circle oppositely. This does not identify the solutions having the same initial velocity. The conserved-plane proof does not rely on coordinates or a gauge potential.

2 Integral charge and the complete quantum spectrum

Let $L_q \to S^2$ be a Hermitian line bundle with unitary connection of curvature

$$F_\nabla = i\frac{q}{2}\,dA. \tag{3}$$

Its first Chern number is

$$\frac{1}{2\pi i}\int_{S^2} F_\nabla = q.$$

Thus integrality of q is exactly the global bundle boundary; a nonintegral local magnetic charge is not admitted to this quantum model.

Complex line bundles over S^2 are classified by $c_1 \in H^2(S^2;\mathbb{Z}) \cong \mathbb{Z}$. Hence L_q is unitarily isomorphic to the standard degree-q homogeneous monopole bundle. After transport to one bundle, two unitary connections with curvature (3) differ by $i\alpha$, where $d\alpha = 0$. Since $H^1_{\mathrm{dR}}(S^2) = 0$, $\alpha = df$; the gauge $g = e^{if}$ carries one connection to the other. Thus the arbitrary frozen connection is unitarily gauge equivalent to the standard homogeneous one.

On $C^\infty(S^2, L_q) = C_c^\infty(S^2, L_q)$ set

$$\mathfrak{q}_q[s] = \int_{S^2} |\nabla s|^2 \, dA,$$

and let Δ_q be its Friedrichs realization. The symmetric Laplace-type expression $\nabla^*\nabla$ on a compact manifold without boundary is essentially self-adjoint on smooth sections. Its unique self-adjoint closure therefore equals this positive Friedrichs operator; the resolvent is compact. Gauge-equivalent connections give unitarily conjugate operators.

Theorem 2 (All monopole-harmonic levels)**.** *The complete spectrum is*

$$\lambda_{n,q} = n(n + |q| + 1) + \frac{|q|}{2}, \qquad n = 0, 1, \ldots, \tag{4}$$

and the multiplicity of $\lambda_{n,q}$ is $2n + |q| + 1$. Hence, for $t > 0$,

$$\operatorname{Tr} e^{-t\Delta_q} = \sum_{n=0}^{\infty} (2n + |q| + 1) e^{-t\lambda_{n,q}}. \tag{5}$$

The degree $-q$ bundle is the complex conjugate of L_q and has the same spectrum.

Proof. It now suffices to use the standard homogeneous connection. Its sections are degree-q equivariant functions on $SU(2)$. Peter–Weyl decomposition contains exactly one copy of each spin $\ell = |q|/2 + n$, $n \geq 0$. The horizontal Laplacian is the total Casimir minus the fixed vertical weight square; it therefore acts by

$$\ell(\ell+1) - \frac{q^2}{4} = n(n + |q| + 1) + \frac{|q|}{2}.$$

The spin space has dimension $2\ell + 1 = 2n + |q| + 1$. Peter–Weyl completeness excludes further levels. The operator realization above and quadratic eigenvalue growth justify the absolutely convergent heat trace. Complex conjugation changes q to $-q$, while (4) depends only on $|q|$. $\square$

At $q = 0$, formula (4) becomes the ordinary spherical harmonic spectrum $n(n+1)$ with multiplicity $2n+1$. At $n = 0$, the lowest monopole level has energy $|q|/2$ and multiplicity $|q| + 1$.

Source lineage

This paper is a source-local reconstruction and makes no priority claim.

References

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